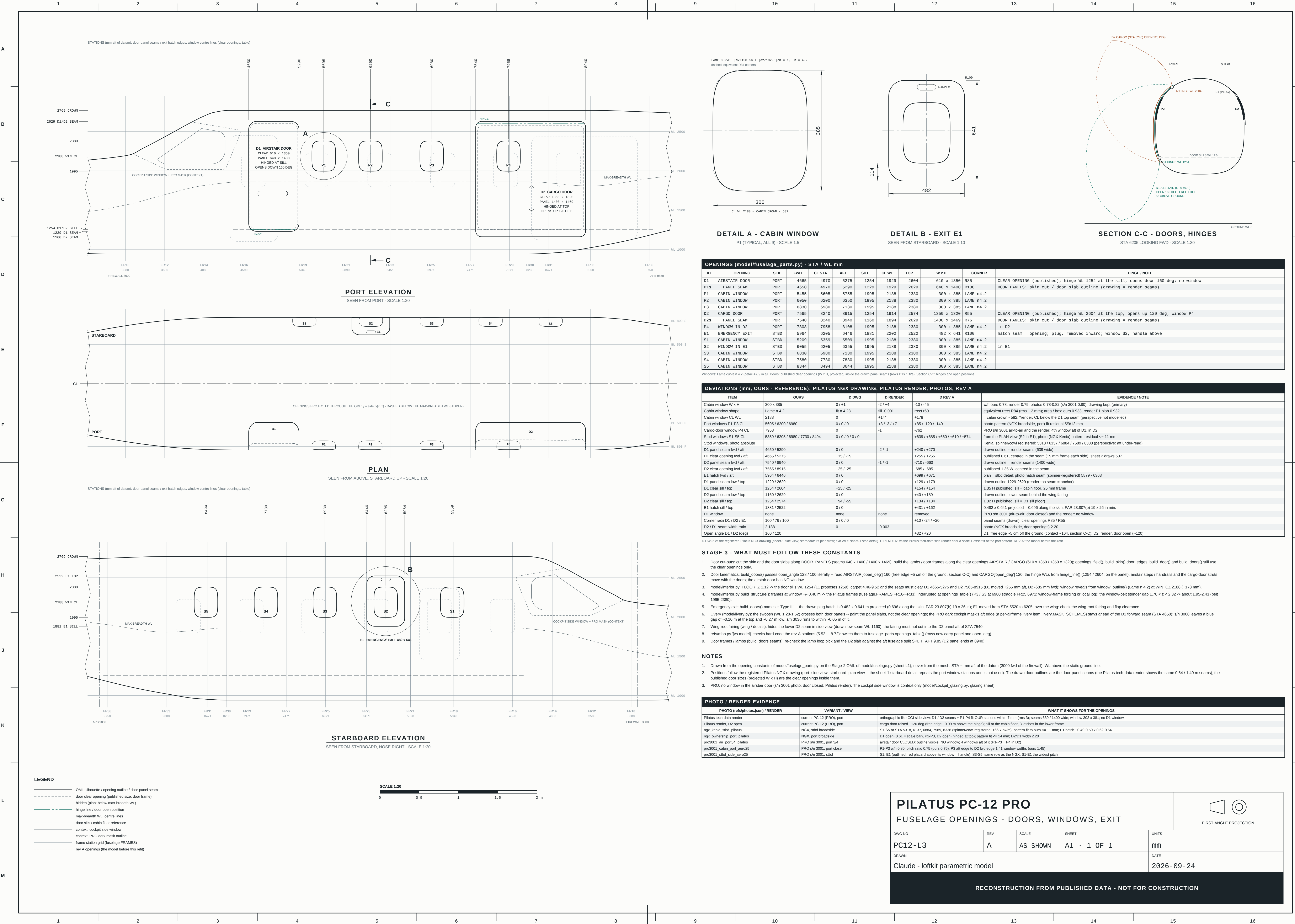
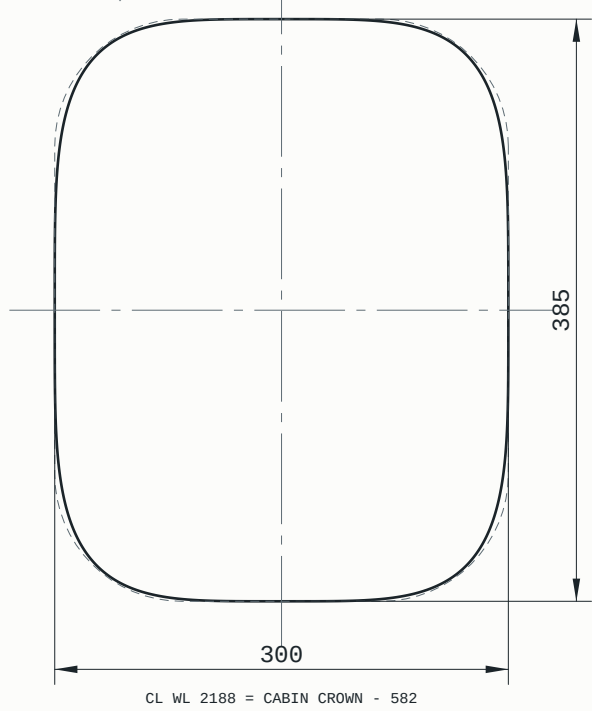


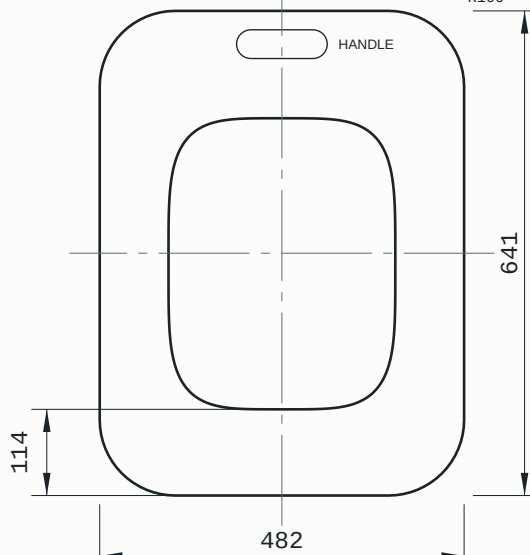


LAME CURVE (dx/350)*h + (dz/192.5)*h = 1, n = 4.2
(squared equivalent R84 corners)



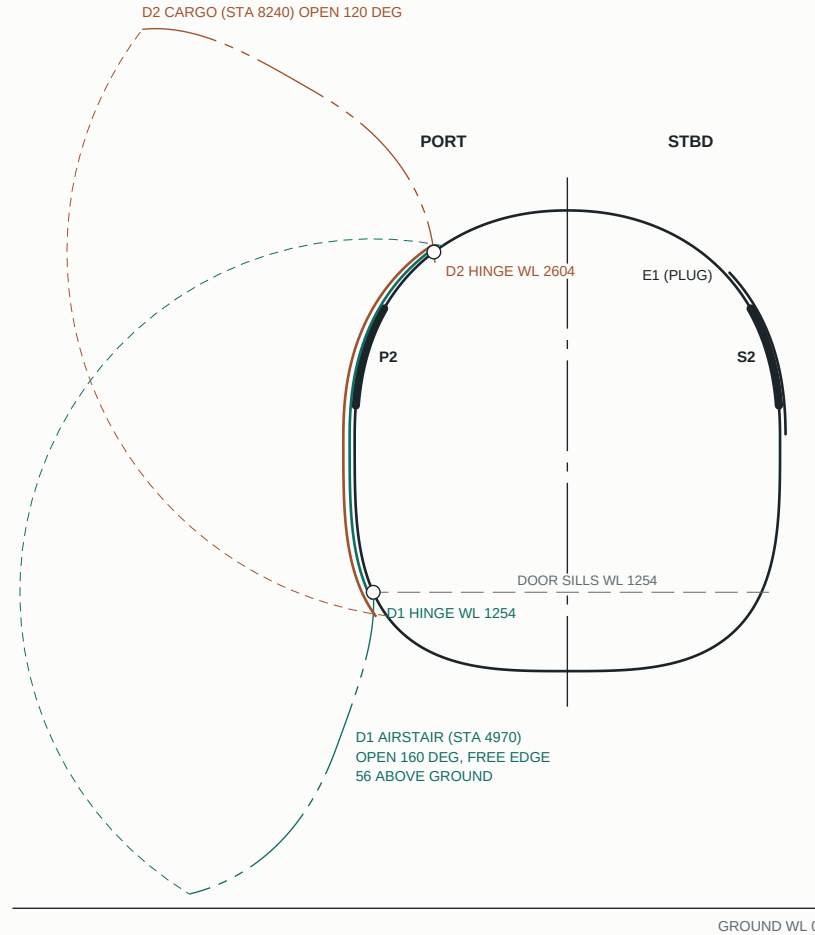
DETAIL A - CABIN WINDOW

P1 (TYPICAL, ALL 9) - SCALE 1:5



DETAIL B - EXIT E1

SEEN FROM STARBOARD - SCALE 1:10



SECTION C-C - DOORS, HINGES

STA 6205 LOOKING FWD - SCALE 1:30

OPENINGS (model/fuselage_parts.py) - STA / WL mm											
ID	OPENING	SIDE	FWD	CL STA	AFT	SILL	CL WL	TOP	W x H	CORNER	HINGE / NOTE
D1	AIRSTAIR DOOR	PORT	4665	4970	5275	1254	1929	2604	610 x 1350	R85	CLEAR OPENING (published); hinge WL 1254 at the sill, opens down 160 deg; no window
D1S	PANEL SEAM	PORT	4650	4970	5290	1229	1929	2629	640 x 1400	R100	DOOR_PANELS: skin cut / door slab outline (drawing = render seams)
P1	CABIN WINDOW	PORT	5455	5695	5755	1995	2188	2380	300 x 385	LAME n4.2	
P2	CABIN WINDOW	PORT	6050	6290	6350	1995	2188	2380	300 x 385	LAME n4.2	
P3	CABIN WINDOW	PORT	6830	6990	7130	1995	2188	2380	300 x 385	LAME n4.2	
D2	CARGO DOOR	PORT	7565	8240	8915	1254	1914	2574	1350 x 1320	R55	CLEAR OPENING (published); hinge WL 2604 at the top, opens up 120 deg; window P4
D2S	PANEL SEAM	PORT	7540	8240	8940	1160	1894	2629	1400 x 1460	R76	DOOR_PANELS: skin cut / door slab outline (drawing = render seams)
P4	WINDOW IN D2	PORT	7800	7950	8100	1995	2188	2380	300 x 385	LAME n4.2	in D2
E1	EMERGENCY EXIT	STBD	5964	6205	6446	1881	2202	2522	482 x 641	R100	hatch seam = opening; plug, removed inward; window S2, handle above
S1	CABIN WINDOW	STBD	5200	5350	5500	1995	2188	2380	300 x 385	LAME n4.2	
S2	WINDOW IN E1	STBD	6055	6205	6355	1995	2188	2380	300 x 385	LAME n4.2	in E1
S3	CABIN WINDOW	STBD	6830	6980	7130	1995	2188	2380	300 x 385	LAME n4.2	
S4	CABIN WINDOW	STBD	7580	7730	7880	1995	2188	2380	300 x 385	LAME n4.2	
S5	CABIN WINDOW	STBD	8344	8494	8644	1995	2188	2380	300 x 385	LAME n4.2	

Windows: Lame curve n 4.2 (detail A), 9 in all. Doors: published clear openings (W x H, projected) inside the drawn panel seams (rows D1S / D2S). Section C-C: hinges and open positions.

DEVIATIONS (mm, OURS - REFERENCE): PILATUS NGX DRAWING, PILATUS RENDER, PHOTOS, REV A						EVIDENCE / NOTE
ITEM	OURS	D DWG	D RENDER	D REV A		
Cabin window W x H	300 x 385	0 / +1	-2 / +4	-10 / +5		with ours 0.78, render 0.79, photos 0.78-0.82 (s/n 3001 0.80); drawing kept (primary)
Cabin window shape	Lame n 4.2	fit n 4.23	fill -0.001	rect #60		equivalent rect R84 (rms 1.2 mm); area / box: ours 0.933, render P1 blob 0.932
Cabin window CL WL	2188	0	+14*	+178		= cabin crown - 582; *render: CL below the D1 top seam (perspective not modelled)
Port windows P1-P3 CL	5605 / 6200 / 6980	0 / 0 / 0	+3 / -3 / +7	+85 / -120 / -140		photo pattern (NGX broadside, port) fit residual 5/9/12 mm
Cargo-door window P4 CL	7958	0	-1	-762		PRO s/n 3001 air-to-air and the render: 4th window aft of D1, in D2
Sbd windows S1-S5 CL	5359 / 6205 / 6980 / 7730 / 8494	0 / 0 / 0 / 0 / 0		+639 / +685 / +660 / +610 / +574		from the PLAN view (S2 in E1); photo (NGX Kenya) pattern residual <= 11 mm
Sbd windows, photo absolute						Kenia, spinner/cowl registered: 5318 / 6137 / 6884 / 7589 / 8338 (perspective: alt under-read)
D1 panel seam fwd / aft	4650 / 5290	0 / 0	-2 / -1	+240 / +270		drawn outline = render seams (639 wide)
D1 clear opening fwd / aft	4655 / 5275	+15 / -15		+255 / +255		published 0.61, centred in the seam (15 mm frame each side), sheet 2 draws 007
D2 panel seam fwd / aft	7540 / 8040	0 / 0	-1 / -1	-710 / -690		drawn outline = render seams (1400 wide)
D2 clear opening fwd / aft	7565 / 8915	+25 / -25		-685 / -685		published 1.35 W, centred in the seam
E1 hatch fwd / aft	5964 / 6446	0 / 0		+699 / +671		plan = sbd detail; photo hatch seam (spinner-registered) 5879 - 6368
D1 panel seam low / top	1229 / 2629	0 / 0		+129 / +179		drawn outline 1229-2629 (render top seam = anchor)
D1 clear sill / top	1254 / 2604	+25 / -25		+154 / +154		1.35 H published; sill = cabin floor, 25 mm frame
D2 panel seam low / top	1160 / 2629	0 / 0		+40 / +189		drawn outline; lower seam behind the wing fairing
D2 clear sill / top	1254 / 2574	+94 / -55		+134 / +134		1.32 H published; sill = D1 sill (floor)
E1 hatch sill / top	1881 / 2522	0 / 0		+431 / +162		0.482 x 0.641 projected = 0.696 along the skin: FAR 23.807(b) 19 x 26 in min.
D1 window	none	none	none	removed		PRO s/n 3001 (air-to-air, door closed) and the render: no window
Corner radii D1 / D2 / E1	100 / 76 / 100	0 / 0 / 0		+10 / -24 / +20		panel seams (drawn); clear openings R85 / R55
D2 / D1 seam width ratio	2.188	0	-0.003			photo (NGX broadside, door openings) 2.20
Open angle D1 / D2 (deg)	160 / 120			+32 / +20		D1: free edge -5 cm off the ground (contact -164, section C-C); D2: render, door open (-120)

D DWG: vs the registered Pilatus NGX drawing (sheet-1 side view, starboard: its plan view, exit WLs: sheet-1 sbd detail). D RENDER: vs the Pilatus tech-data side render after a scale + offset fit of the port pattern. REV A: the model before this ref.

STAGE 3 - WHAT MUST FOLLOW THESE CONSTANTS

- Door cut-outs: cut the skin and the door slabs along DOOR_PANELS (seams 640 x 1400 / 1400 x 1460), build the jambs / door frames along the clear openings AIRSTAIR / CARGO (610 x 1350 / 1350 x 1320); openings_field(), build_skin() door_edges, build_door() and build_doors() still use the clear openings only.
- Door kinematics: build_doors() passes open_angle 128 / 100 literally -- read AIRSTAIR[open_deg] 160 (free edge ~5 cm off the ground, section C-C) and CARGO[open_deg] 120, the hinge WLs from hinge_line() (1254 / 2604, on the panel); airstair steps / handrails and the cargo-door struts move with the doors; the airstair door has NO window.
- modelInterior.py: FLOOR_Z 1.12 -> the door sills WL 1254 (L1 proposes 1259); carpet 4.46-9.52 and the seats must clear D1 4665-5275 and D2 7565-8915 (D1 moved +255 mm aft, D2 -685 mm fwd); window reveals from window_outline() (Lame n 4.2) at WIN_CZ 2188 (+178 mm).
- modelInterior.py build_structure(): frames at window +/- 0.40 m -> the Pilatus frames (fuselage.FRAMES FR16-FR33), interrupted at openings_table() (P3 / S3 at 6980 straddle FR25 6971: window-frame forging or local jog); the window-belt stringer gap 1.70 < z < 2.32 -> about 1.95-2.43 (belt 1995-2380).
- Emergency exit: build_doors() names it 'Type III' -- the drawn plug hatch is 0.482 x 0.641 m projected (0.696 along the skin, FAR 23.807(b) 19 x 26 in); E1 moved from STA 5520 to 6205, over the wing: check the wing-root fairing and flap clearance.
- Livery (model/livery.py): the swoosh (WL 1.28-1.52) crosses both door panels -- paint the panel slabs, not the clear openings; the PRO dark cockpit mask's aft edge (a per-airframe livery item, livery.MASK_SCHEMES) stays ahead of the D1 forward seam (STA 4650); s/n 3008 leaves a blue gap of ~0.10 m at the top and ~0.27 m low, s/n 3036 runs to within ~0.05 m of it.
- Wing-root fairing (wing / details): hides the lower D2 seam in side view (drawn low seam WL 1160); the fairing must not cut into the D2 panel aft of STA 7540.
- refs/mbp.py [vs model]: checks hard-code the rev-A stations (5.52 ... 8.72); switch them to fuselage_parts.openings_table() (rows now carry panel and open, deg).
- Door frames / jambs (build_doors seams): re-check the jamb loop pick and the D2 slab against the aft fuselage split SPLIT_AFT 9.85 (D2 panel ends at 8940).

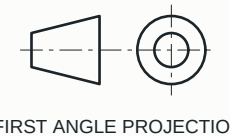
NOTES

- Drawn from the opening constants of model/fuselage_parts.py on the Stage-2 OML of model/fuselage.py (sheet L1), never from the mesh. STA = mm aft of the datum (3000 fwd of the firewall); WL above the static ground line.
- Positions follow the registered Pilatus NGX drawing (port: side view; starboard: plan view -- the sheet-1 starboard detail repeats the port window stations and is not used). The drawn door outlines are the door-panel seams (the Pilatus tech-data render shows the same 0.64 / 1.40 m seams); the published door sizes (projected W x H) are the clear openings inside them.
- PRO: no window in the airstair door (s/n 3001 photo, door closed; Pilatus render). The cockpit side window is context only (model/cockpit_glazing.py, glazing sheet).

PHOTO / RENDER EVIDENCE		
PHOTO (refs/photos.json) / RENDER	VARIANT / VIEW	WHAT IT SHOWS FOR THE OPENINGS
Pilatus tech-data render	current PC-12 (PRO), port	orthographic-like CGI side view: D1 / D2 seams + P1-P4 fit OUR stations within 7 mm (rms 3); seams 639 / 1400 wide; window 302 x 361; no D1 window
Pilatus render: D2 open	current PC-12 (PRO), port	cargo door raised ~120 deg (free edge ~0.90 m above the hinge); sill at the cabin floor; 3 latches in the lower frame
ngx_kenya_sbd_pilatus	NGX, sbd broadside	S1-S5 at STA 5318, 6137, 6884, 7580, 8338 (spinner/cowl registered, 166.7 pphm); pattern fit to ours <= 11 mm; E1 hatch ~0.49-0.50 x 0.62-0.64
ngx_ownership_port_pilatus	NGX, port broadside	D1 open (0.61 = scale bar); P1-P3, D2 open (hinged at top); pattern fit <= 14 mm; D2/D1 width 2.20
pro3001_ar_port34_pilatus	PRO s/n 3001, port 3/4	airstair door CLOSED: outline visible. No window, 4 windows aft of it (P1-P3 + P4 in D2)
pro3001_cabin_port_aero25	PRO s/n 3001, port close	P1-P3 with 0.80, pitch ratio 0.75 (ours 0.76); P3 aft edge to D2 fwd edge 1.41 window widths (ours 1.45)
pro3001_sbd_side_aero25	PRO s/n 3001, sbd	S1, E1 (outlined, red placard above its window = handle); S3-S5: same row as the NGX. S1-E1 the widest pitch

PILATUS PC-12 PRO

FUSELAGE OPENINGS - DOORS, WINDOWS, EXIT



FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L3	A	AS SHOWN	A1 · 1 OF 1	mm

DRAWN	DATE
Claude - loftkit parametric model	2026-09-24

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION